SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	27.10.26	**Duration:**	60 minutes

Code of Conduct

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description

- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

(15%)	sections.			components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

1. Study and Analyze the Scenario

1.1 Business Domain

The system belongs to the IT service management and internal technical support domain. Organizations depend on IT infrastructure, applications, networks and computing devices for daily work. When employees face technical problems, the IT team needs a structured way to receive, prioritize, assign and resolve those problems. The proposed platform provides one place for managing the complete service-ticket lifecycle.

1.2 Scenario Analysis

The current problem is that employees may report technical issues through email or phone. These channels do not always provide a common record for prioritization and tracking. A centralized help desk platform can convert each reported issue into a service ticket, give it a category and priority, assign it to the appropriate support team and maintain its status until resolution.

The platform also needs a technical knowledge base so that common problems and their solutions can be reused. Service-level monitoring is included so that support staff and managers can identify tickets that are close to or beyond their agreed response or resolution target. User feedback and service analytics provide a way to evaluate the quality of IT support.

1.3 Stakeholders

Stakeholder	Role and Interest
Employees / End Users	Report IT issues, follow ticket progress, search common solutions and provide feedback.
IT Support Staff	Receive assigned tickets, investigate issues, update ticket status and record resolution details.
IT Manager	Monitor ticket workload, SLA performance, resolution trends and overall service quality.
System Administrator	Manage users, roles, teams, categories, SLA settings and knowledge-base access.
Organization / Management	Expect efficient IT support, reduced downtime and better visibility of service performance.

1.4 Existing Challenges

- Reports are spread across email and phone conversations instead of being maintained in one ticket record.
- Important incidents may not always be prioritized consistently.
- Manual assignment can delay sending a problem to the correct support team.
- Employees may not know whether their issue has been received, assigned or resolved.
- The same common technical problems may be solved repeatedly without a reusable knowledge record.
- SLA deadlines can be missed when response and resolution times are not monitored systematically.
- Managers may not have a clear view of ticket volume, priority, resolution time and recurring problems.

1.5 System Objectives

- Create IT service tickets for employee-reported problems.
- Categorize technical problems into suitable service areas.
- Prioritize incidents according to their impact and urgency.
- Assign tickets to appropriate support teams or support staff.
- Track ticket status throughout the service lifecycle.
- Monitor service-level agreements and identify tickets at risk of delay.
- Maintain a technical knowledge base for common problems and solutions.
- Generate IT service analytics for service-performance review.

1.6 Expected Outcomes

- Faster IT issue resolution.
- Improved service desk efficiency.
- Better ticket prioritization.
- Reduced repeated support requests.
- Improved SLA compliance.
- Better visibility into IT service performance.

2. Functional and Non-Functional Requirements

2.1 Functional Requirements

ID	Function	Requirement
FR-01	User Authentication	The system shall allow registered users to log in using valid credentials.
FR-02	Create Ticket	The system shall allow an employee to create an IT issue ticket with title, description, category and relevant details.
FR-03	Unique Ticket ID	The system shall generate a unique ticket ID for every new ticket.
FR-04	Categorize Ticket	The system shall classify tickets into suitable categories such as hardware, software, network or access.
FR-05	Prioritize Incident	The system shall assign a priority using the impact and urgency of the reported issue.
FR-06	Assign Ticket	The system shall assign a ticket to an appropriate IT support team or authorized support staff member.
FR-07	Update Ticket Status	Authorized support staff shall update the ticket through statuses such as Open, Assigned, In Progress, Resolved and Closed.
FR-08	Track Ticket	Employees shall be able to view the current status and basic progress of their own tickets.
FR-09	SLA Monitoring	The system shall record applicable SLA targets and indicate tickets approaching or exceeding the target.
FR-10	Knowledge Base Search	Users shall be able to search technical knowledge-base articles using keywords.
FR-11	Manage Knowledge Base	Authorized support users shall add, edit and remove knowledge-base articles.
FR-12	User Feedback	Employees shall be able to provide a rating and short comment after a ticket is resolved.
FR-13	Service Analytics	The system shall provide ticket counts, status distribution, priority distribution and resolution-time information.
FR-14	Notifications	The system shall notify relevant users about important ticket events such as assignment, status change and resolution.
FR-15	Role-Based Access	The system shall provide functions according to the user's role and restrict unauthorized operations.

2.2 Non-Functional Requirements

ID	Attribute	Requirement
NFR-01	Performance	Common ticket pages and searches should respond within a few seconds under expected organizational usage.
NFR-02	Security	Authentication and role-based access shall protect ticket and administrative information.
NFR-03	Reliability	Ticket creation and updates should be stored correctly without data loss during normal operation.
NFR-04	Usability	The interface should be simple enough for an employee with basic computer knowledge to report an issue.
NFR-05	Scalability	The design should support growth in users, tickets and support teams without major redesign.

ID	Attribute	Requirement
NFR-06	Maintainability	The platform should use modular components so categories, SLA rules and knowledge articles can be changed easily.
NFR-07	Availability	The platform should be available during normal organizational working hours with minimum planned downtime.
NFR-08	Auditability	Important actions such as assignment, status changes and resolution updates should be traceable.

3. Actors and Use Case Modelling

3.1 Actors

Actor	Type	Main Responsibility
Employee	Primary	Creates and tracks tickets, searches knowledge articles and provides feedback.
IT Support Staff	Primary	Works on assigned tickets, updates status, records resolution and uses knowledge articles.
IT Manager	Primary	Monitors SLA performance, workload and IT service analytics.
System Administrator	Secondary	Manages users, roles, categories, teams, SLA configuration and knowledge-base administration.

3.2 Major Use Cases

Actor	Major Use Cases
Employee	Login; Create Ticket; Track Ticket; Search Knowledge Base; Provide Feedback
IT Support Staff	Login; View Assigned Tickets; Categorize/Prioritize Ticket; Update Status; Record Resolution; Search Knowledge Base
IT Manager	Login; Monitor SLA; View IT Analytics; Review Service Performance
System Administrator	Login; Manage Users/Roles; Manage Categories/Teams; Configure SLA Rules; Manage Knowledge Base

3.3 UML Use Case Diagram

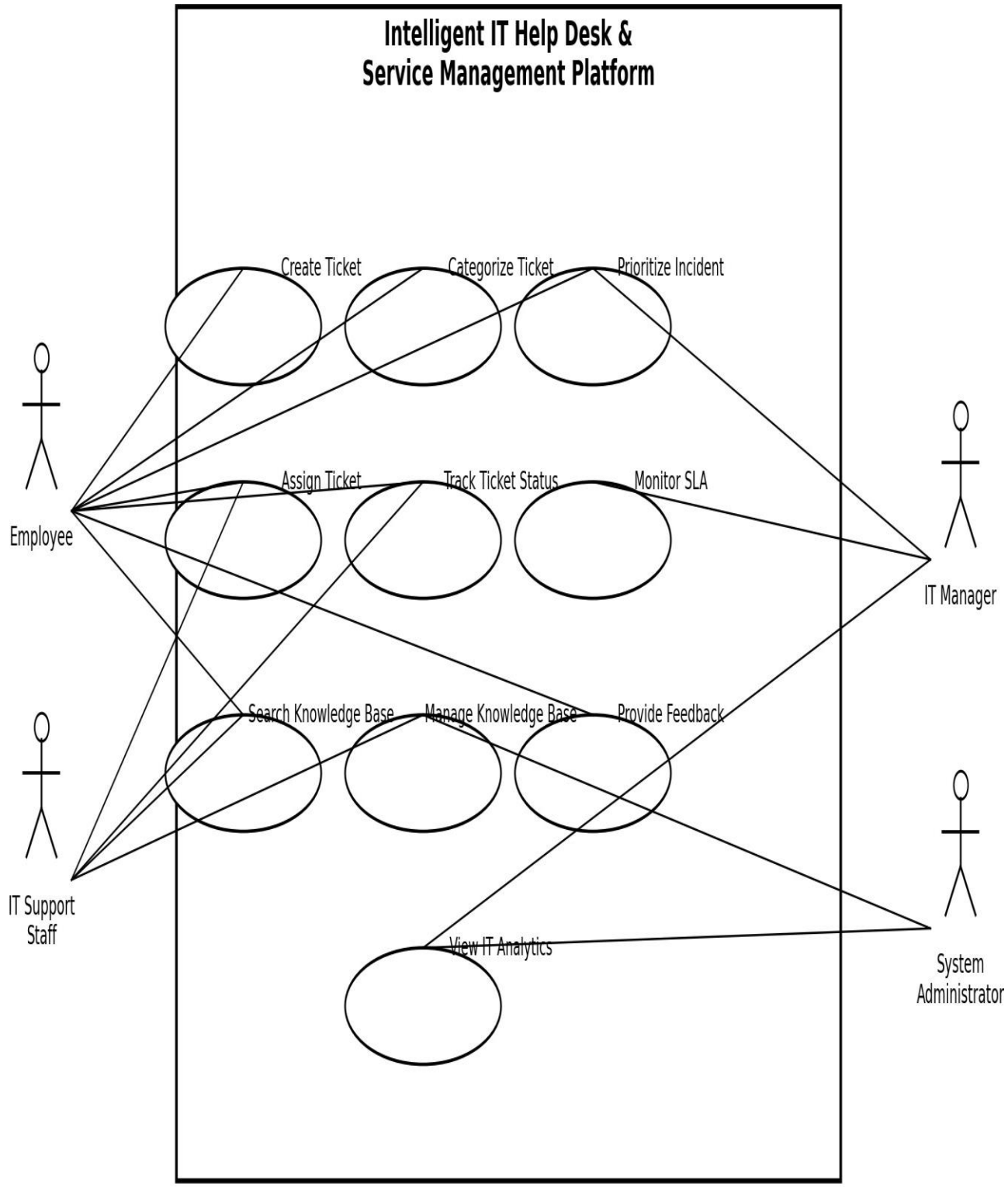


Figure 1. UML Use Case Diagram

Figure 1. UML Use Case Diagram

The diagram represents the main interaction between the four actors and the proposed platform. The employee initiates service requests and follows them, support staff process the requests, the IT manager monitors service performance, and the system administrator maintains system configuration and knowledge-base functions.

4. Software Requirement Specification (SRS)

4.1 Introduction

The Intelligent IT Help Desk and Service Management Platform is a centralized software system for managing internal IT service requests. It provides a common workflow from issue reporting to ticket closure. The system is intended to reduce the difficulties caused by scattered email and phone-based support and to improve visibility for both employees and IT management.

4.2 Purpose

The purpose of this SRS is to define the functional behaviour, quality requirements, users, use cases, data requirements, interfaces and limitations of the proposed system. It provides a clear reference for a future design and implementation team.

4.3 Problem Description

Organizations depend heavily on IT infrastructure, applications, networks and computing devices. Employees frequently experience technical issues, but email- and phone-based support can make ticket prioritization and tracking difficult. Without a centralized service-management process, issues may be delayed, duplicated or assigned incorrectly. The proposed platform addresses these problems by providing structured ticket creation, categorization, prioritization, assignment, tracking, SLA monitoring, knowledge management, feedback and analytics.

4.4 Scope of the System

The system covers internal IT incident and service-ticket management. It includes user authentication, ticket creation, categorization, prioritization, assignment, status tracking, SLA monitoring, knowledge-base

management, feedback collection, notifications and IT service analytics.

The minimum scope does not include physically repairing hardware, directly controlling network devices, purchasing IT assets or replacing the organization's existing enterprise monitoring tools. The platform records and manages the service process followed by the IT team.

4.5 System Features

Feature	Description
Ticket Management	Create, identify, assign, update and close IT service tickets.
Categorization and Prioritization	Classify technical issues and assign priority according to impact and urgency.
Service Lifecycle Tracking	Maintain ticket status from creation through assignment, work, resolution and closure.
SLA Monitoring	Compare ticket progress with configured response and resolution targets.
Knowledge Base	Store and search solutions for common technical problems.
User Feedback	Collect a rating and comment after a ticket is resolved.
Service Analytics	Provide useful statistics for reviewing IT service performance.
Role-Based Access	Give employees, support staff, managers and administrators suitable permissions.

4.6 Detailed Use Case Descriptions

ID	Use Case	Actor	Precondition	Main Flow	Postcondition	Exception
UC-01	Create IT Ticket	Employee	User is logged in.	Enter issue title and description; select/confirm category; submit ticket.	System creates the ticket and displays a unique ticket ID.	Required information is missing.
UC-02	Handle Assigned Ticket	IT Support Staff	Ticket is assigned to the staff member/team.	Open ticket; review details; diagnose issue; update status; enter resolution.	Ticket contains current status and resolution details.	Ticket is unavailable or user lacks permission.
UC-03	Track Ticket	Employee	Employee has an existing ticket.	Open My Tickets; select ticket; view current status and updates.	Latest ticket status is displayed.	Ticket does not belong to the employee.
UC-04	Monitor SLA	IT Manager	SLA rules and ticket dates are available.	Open SLA view; review approaching/overdue tickets; examine performance.	SLA status and service performance are displayed.	SLA information is incomplete.
UC-05	Search Knowledge Base	Employee / Support Staff	Knowledge articles exist.	Enter keyword; search; open relevant article.	Matching articles are displayed.	No matching article is found.
UC-06	Provide Feedback	Employee	Ticket is resolved.	Open resolved ticket; select rating; enter optional comment; submit.	Feedback is stored against the ticket.	Ticket is not resolved.
UC-07	View IT Analytics	IT Manager	Manager access is available.	Open analytics; select period/category; review statistics.	Requested service statistics are displayed.	User lacks manager access.

4.7 Data Requirements

Entity	Important Data Fields
User	User ID, name, email, department, role, account status
Ticket	Ticket ID, title, description, category, priority, status, created date/time, updated date/time
Assignment	Ticket ID, support team, assigned staff, assignment date/time
SLA	SLA type, target response time, target resolution time, due time, current SLA status
Knowledge Article	Article ID, title, problem description, solution, category, last updated date
Feedback	Feedback ID, ticket ID, rating, comment, submitted date

Entity	Important Data Fields
Ticket Activity / Audit	Ticket ID, action, previous value, new value, user, date/time

4.8 External Interface Requirements

Interface	Requirement
User Interface	A web-based interface shall provide login, ticket creation, ticket list, ticket details, knowledge search and dashboards.
Input Interface	Forms shall accept ticket details, category/priority information, status updates and feedback.
Output Interface	The system shall display ticket status, notifications, SLA indicators, knowledge articles and analytics.
Database Interface	The application shall store user, ticket, assignment, SLA, knowledge-base and feedback information in a structured database.
Notification Interface	The system may provide in-application or email notifications for important ticket events.

4.9 System Workflow

The expected service workflow is: Employee reports an issue → system creates a ticket → ticket is categorized → priority is determined → ticket is assigned → support staff work on the issue → status is updated → SLA is monitored during the process → support staff record the resolution → ticket is marked Resolved/Closed → employee provides feedback → service data becomes available for analytics.

4.10 Security Requirements

- Only authenticated users shall access protected functions.
- Employees shall be able to view their own tickets and shall not modify another employee's ticket without authorization.
- Support staff shall access tickets according to their assigned role/team permissions.
- Manager and administrator functions shall be restricted to authorized accounts
- Important ticket changes should be recorded for accountability.

5. Assumptions and Constraints

5.1 Assumptions

- Each employee and IT staff member has a unique system account.
- The organization has defined IT support teams and service categories.
- The organization defines SLA targets for different service priorities or categories.
- Users have access to the organizational network or approved internet connection.
- Employees provide reasonably accurate descriptions of their technical issues.
- Authorized support staff review knowledge-base articles before they are treated as official solutions.
- IT support staff update ticket status as work progresses.

5.2 Technical Constraints

The platform depends on a web application, a database and network connectivity. The design should be practical for the organization's available infrastructure and should not assume unlimited computing or storage resources.

5.3 Operational Constraints

The quality of categorization and assignment depends partly on the information supplied by employees and the configuration maintained by the IT team. Support staff must also keep ticket status and resolution information up

6. Requirement Traceability Summary

Problem / Objective	Requirement(s)	Expected Benefit
Scattered IT issue reporting	FR-02, FR-03	Centralized and identifiable tickets
Difficulty in categorization	FR-04	Issues reach the appropriate service area
Poor prioritization	FR-05	Important incidents receive suitable attention
Manual ticket assignment	FR-06	Tickets reach the correct support team
Lack of progress visibility	FR-07, FR-08	Employees and staff can follow ticket lifecycle
SLA monitoring difficulty	FR-09	At-risk and overdue tickets become visible
Repeated common problems	FR-10, FR-11	Solutions can be reused through the knowledge base
Need for user feedback	FR-12	Service quality can be reviewed from user perspective
Limited service visibility	FR-13	Managers receive useful IT service statistics

7. Conclusion

The Intelligent IT Help Desk and Service Management Platform provides a structured approach to internal IT service management. The analysis identifies the business domain, stakeholders, challenges, objectives and expected outcomes. The functional and non-functional requirements define the behaviour and quality expected from the platform. Actors and use cases describe how employees, support staff, managers and administrators interact with the system.

The SRS further defines the system scope, features, data requirements, external interfaces, use cases, assumptions and constraints. Requirement validation shows that the major objectives in the assigned problem statement have

been addressed. The proposed system therefore provides a practical foundation for a future implementation while keeping the present assessment focused on requirements analysis rather than software development.

8. Quick Viva / Explanation Points

If asked to explain the document, the system can be described in simple terms as follows:

- The main problem is that IT complaints through email and phone are difficult to prioritize and track.
- The solution is a centralized IT help desk platform where every complaint becomes a ticket.
- The four main actors are Employee, IT Support Staff, IT Manager and System Administrator.
- The important workflow is Create → Categorize → Prioritize → Assign → Work → Resolve/Close → Feedback.
- SLA monitoring checks whether tickets are being handled within the agreed service time.
- The knowledge base reduces repeated support work by keeping solutions for common problems.
- Analytics gives the IT manager visibility into ticket volume, priorities, resolution time and service performance.

The current assessment is requirement analysis and SRS preparation; coding is not required by the supplied assessment instructions. to date.

8.1 Economic Constraints

The solution should be practical within the organization's IT budget. Unnecessary licensing, infrastructure and maintenance costs should be avoided when equivalent capabilities can be provided using the organization's existing resources.

8.2 Security and Legal Constraints

Ticket information may contain internal organizational details. Access must therefore be limited according to user roles, and administrative or service information should not be exposed to unauthorized users. The organization is responsible for defining its own data-retention and access policies.

8.3 Time Constraint

This assessment is focused on requirement analysis and SRS preparation. A complete production implementation, deployment and integration with all existing organizational systems are outside the scope of the present assessment.

9. Requirement Completeness and Consistency Validation

9.1 Stakeholder Coverage Validation

The identified requirements cover the main stakeholder needs. Employees can report and track issues and provide feedback. IT support staff can receive, work on and resolve tickets. IT managers can monitor SLA performance and service analytics. System administrators can manage users, roles, categories, teams and knowledge-base configuration.

9.2 Requirement Completeness Check

Scenario Need	Status	Supporting Requirements
Create IT service tickets	Covered	FR-02, FR-03
Categorize technical problems	Covered	FR-04
Prioritize incidents	Covered	FR-05
Assign tickets to support teams	Covered	FR-06
Track ticket status	Covered	FR-07, FR-08
Monitor service-level agreements	Covered	FR-09
Maintain technical knowledge base	Covered	FR-10, FR-11
Collect user feedback	Covered	FR-12
Generate IT service analytics	Covered	FR-13

9.3 Consistency Check

- The same ticket lifecycle terms are used throughout the SRS: Open, Assigned, In Progress, Resolved and Closed.
- Employee access is separated from support and administrative access through role-based permissions.
- Priority is described using impact and urgency rather than an undefined label such as 'important'.
- SLA monitoring is treated as a service-management function and does not claim to perform the actual technical repair.
- The analytics requirements are limited to ticket and service-performance information described by the scenario.
-